Day 2 Student Handout

Geometry Summer Credit | Slope, Lines, and Transversals

Name:	Date:	Period:

Learning Targets

I can calculate slope from two points and use slope to identify parallel and perpendicular lines. I can use angle relationships formed by transversals to write and solve equations.

Lesson 2A: Slope and Line Relationships

Warm-Up

1.	Find the midpoint of A(-6, 8) and B(2, -4). Then find the distance between the points.

2.	Solve each equation: $3x + 12 = 90$ and $5x - 7 = 2x + 23$.

Concept Launch

Notice and Wonder

Slope measures steepness: the ratio of vertical change to horizontal change. Lines with the same slope never meet; lines with opposite reciprocal slopes meet at right angles.

3.	Plot P(-4, -1), Q(2, 3), R(-3, 4), and S(3, 8). Compare the slope of PQ and RS. What do you notice?

Guided Examples

Problem	Work / notes
Find the slope through A(-2, 5) and B(4, -1).	Change in y: Change in x: Slope:

Line l has slope 3/4. What slope would be parallel to l? What slope would be perpendicular to l?	Parallel slope: Perpendicular slope:
Are the lines through A(1, 2), B(5, 5) and C(-2, 4), D(2, 7) parallel, perpendicular, or neither?	Slope AB: Slope CD: Conclusion:
A line passes through (2, -3) with slope -2. Find two more points on the line.	Use rise/run: Point 1: Point 2:

Practice

4.	Find the slope through (-5, 2) and (1, 8).

5.	Find the slope through (3, -2) and (7, -2). What kind of line is this?

6.	A line has slope -5/2. Give the slope of a parallel line and the slope of a perpendicular line.

7.	Decide whether the lines with slopes 4/3 and -3/4 are parallel, perpendicular, or neither. Explain.

8.	Error analysis: A student says the slope from (1, 6) to (5, 2) is 1 because both coordinates changed by 4. Explain and correct the error.

Day 2 Student Handout

Lesson 2B | Transversals and Algebra with Angle Pairs

Lesson 2B: Transversals and Angle Relationships

Warm-Up

1.	Two angles form a linear pair. One angle is 68 degrees. Find the other angle.

2.	Vertical angles are labeled $4x + 11$ and $7x - 31$. Find x.

3.	Write one sentence that explains the difference between complementary and supplementary angles.

Key Vocabulary

Term	What it means / sketch
Transversal	
Corresponding angles	
Alternate interior angles	
Alternate exterior angles	
Same-side interior angles	

Concept Launch

Look for Structure

When a transversal crosses parallel lines, matching angle positions are congruent and same-side interior angles are supplementary.

4.	Sketch two parallel horizontal lines cut by a slanted transversal. Label one acute angle 62 degrees. Use angle relationships to label as many other angles as you can.

Guided Examples

Problem	Work / notes
Parallel lines are cut by a transversal. A corresponding angle to 74 degrees is labeled 3x + 8. Find x.	Equation: Solve:
Alternate interior angles are labeled 5x - 12 and 2x + 33. Find x and the angle measure.	Equation: Solve: Angle measure:
Same-side interior angles are labeled 4x + 15 and 3x - 10. Find x and both angle measures.	Equation: Solve: Angle measures:

Practice

5.	A corresponding angle to 118 degrees is labeled 6x - 2. Find x.

6.	Alternate interior angles are labeled 8x + 4 and 5x + 31. Find x.

7.	Same-side interior angles are labeled 2x + 40 and 4x + 20. Find x and both angle measures.

8.	Error analysis: A student sets same-side interior angles equal. Explain the mistake and write the correct relationship.

Reflection

9.	Which topic feels easier today? What question remains?